

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 2: Atmospheric Pressure, Altimetry & Q-Codes

Syllabus Source: CAE Oxford (pp. 33-68) | Pass Target: >= 90% AviationExam

1. PRESSURE GRADIENTS & ALTIMETER Q-CODES

Atmospheric pressure is the weight of the air column above a unit surface area. It governs altimetry:

Q-Code / Datum	Reference Pressure Datum Plane	Altimeter Reading & Cockpit Usage
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QNH (Nautical Height)	Atmospheric pressure reduced to Mean Sea Level (MSL) using ISA lapse rate.	Altimeter reads ALTITUDE above MSL. Reads exact aerodrome elevation on runway!
QFE (Field Elevation)	Actual atmospheric pressure measured at the aerodrome reference datum.	Altimeter reads HEIGHT (AGL) above aerodrome. Reads exactly ZERO on runway threshold!
QNE / Standard (1013.25)	Standard datum plane 1013.25 hPa.	Altimeter reads FLIGHT LEVELS (FL) above Transition Altitude. (e.g. FL 350 = 35,000 ft standard).

2. TEMPERATURE ERRORS IN ALTIMETRY

An altimeter is calibrated strictly to the ISA atmosphere. Non-standard temperatures create dangerous errors:

- Rule of Thumb: 'From High to Low or Hot to Cold, Look Out Below!' When flying into air that is colder than ISA, the atmospheric column compresses. The altimeter over-reads, and your TRUE ALTITUDE IS LOWER than indicated!
- Cold Temperature Correction: True Altitude Correction = 4 ft x (Height above station / 1,000 ft) x (ISA Deviation below standard).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Cold Air Altimeter Correction & P

Rule 1 (High to Low, look out below): - Flying from high pressure (1025 hPa) to low pressure (1005 hPa) without resetting subscale: Error = 20 hPa x 27 ft = 540 ft. - The altimeter overreads: indicated altitude is 5,000 ft, but true altitude is only 4,460 ft! Rule 2 (ICAO Cold Temperature Correction on Final Approach): - Aerodrome temperature is -20 C (ISA -35 C). Minimum safe altitude is 2,000 ft AGL. - Formula: Correction = 4 ft x (Height / 1,000 ft) x Delta ISA cold = 4 x 2 x 35 = 280 ft. - Crucial Action: Pilot must fly at 2,280 ft indicated to ensure true clearance!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Altimeter Reading Changing QNH to Standard

SCENARIO (Classic Altimetry Transition Drill): An aircraft is cruising at an Indicated Altitude of 5,000 ft with the subscale set to QNH 993 hPa. ATC clears the pilot to climb to FL 80 and instructs: 'Set Standard 1013 hPa'. QUESTION: What will the altimeter indicate immediately after setting the subscale to 1013.25 hPa before any climb takes place? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Pressure Difference on the subscale: - New setting = 1013 hPa - Old setting = 993 hPa - Pressure Change = $1013 - 993 = +20$ hPa. Step 2: Understand the physical rule for subscale adjustment: - Turning the subscale to a HIGHER pressure number forces the altimeter hands to WIND UP (indicate a higher altitude)! - Rate of change: 1 hPa = approx 27 ft (or 30 ft in low-altitude rules). - Using standard EASA rule of 27 ft/hPa (or 30 ft/hPa): $* 20 \text{ hPa} \times 27 \text{ ft/hPa} = 540 \text{ ft}$ (using 30 ft/hPa = 600 ft). Step 3: Calculate the New Indicated Altitude: - New Indicated Altitude = Old Indicated + Adjustment - Using 27 ft/hPa: $5,000 \text{ ft} + 540 \text{ ft} = 5,540 \text{ ft}$! - (Using 30 ft/hPa: $5,000 \text{ ft} + 600 \text{ ft} = 5,600 \text{ ft}$). FINAL ANSWER: The altimeter pointer winds clockwise to indicate approximately 5,540 ft to 5,600 ft.

[!] EXAM TRAP (AVIATIONEXAM) - The High to Low Pressure Trap

- If you fly from an area of HIGH pressure into an area of LOW pressure without resetting the altimeter: - The altimeter CONTINUES TO READ THE OLD ALTITUDE, but your airplane has descended! - You are physically CLOSER TO THE GROUND than your instrument displays ('High to low, look out below!').

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: If an aircraft flies from a region of HIGH pressure into an area of LOW pressure without adjusting the altimeter, what will the altimeter indicate?

- [A]: It reads lower than the true altitude
- [B]: It reads HIGHER than the true altitude (the aircraft is lower than the pilot believes)
- [C]: It reads the correct altitude
- [D]: It fails

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The altimeter measures ambient pressure. In low pressure, the altimeter assumes the plane has climbed, so it over-reads. The aircraft's true altitude is lower.

Q2: When on the ground at an aerodrome with the altimeter subscale set to QFE, what does the altimeter indicate?

- [A]: Aerodrome elevation above sea level
- [B]: Exactly ZERO feet
- [C]: Density altitude
- [D]: Pressure altitude

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. QFE is the measured pressure at the airfield datum. With QFE set on the ground, the altimeter indicates height above the field, which is 0 ft.

Q3: What is the Transition Level (TRL)?

- [A]: The highest altitude where QNH is used
- [B]: The LOWEST AVAILABLE FLIGHT LEVEL above the Transition Altitude (TA)
- [C]: 3,000 ft
- [D]: The level of the tropopause

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Transition Altitude (TA) is the highest altitude using QNH. Transition Level (TRL) is the lowest usable Flight Level using standard 1013.25 hPa above the transition layer.